Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

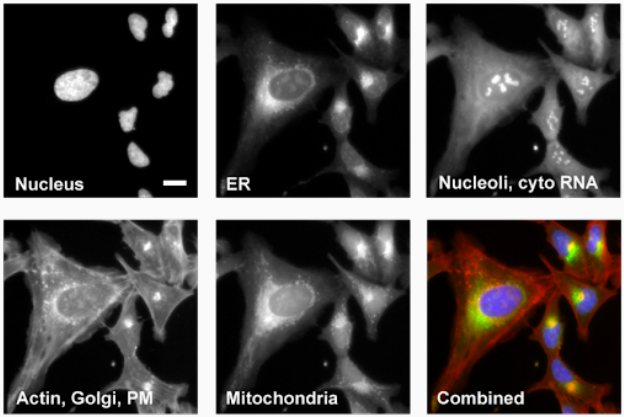
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University

10 November 2026

What is Cell Painting?

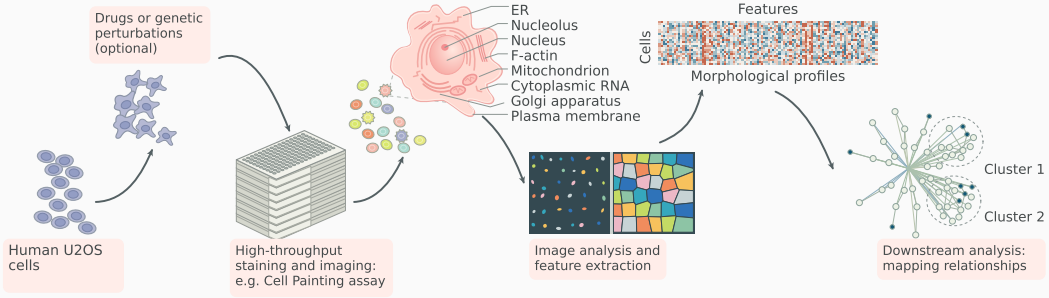
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

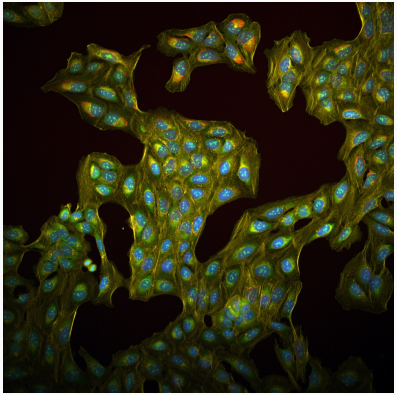
Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

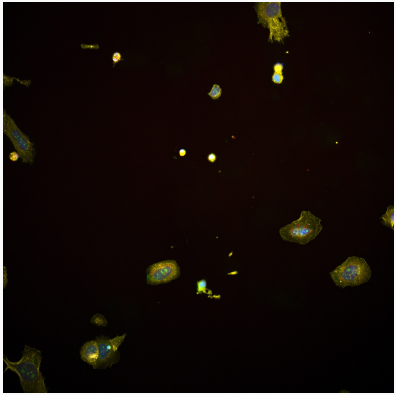
How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

An extreme example: nocodazole

DMSO control



Nocodazole



OPA1 links genes to mitochondrial shape

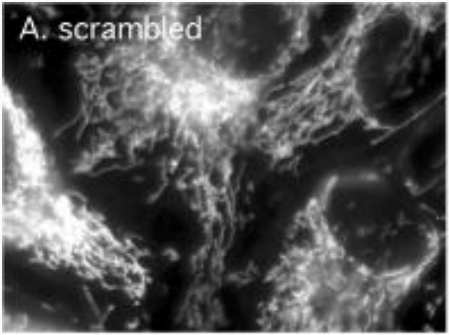
- OPA1 is a dynamin-related GTPase in the mitochondrial inner membrane.
- It supports inner-membrane fusion and mitochondrial network organization.
- OPA1 depletion produces fragmented mitochondrial networks.

A genetic example with a recognizable imaging phenotype.

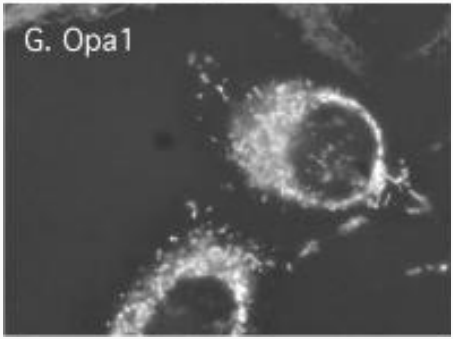
Griparic et al., JCB (2007); Song et al., JCB (2007)

OPA1 depletion fragments mitochondria

Control siRNA



OPA1 siRNA



HeLa cells; MitoTracker; OPA1 depletion for 48 hours.

Griparic et al., JCB (2007), Fig. 3A,G; cropped panels, CC BY-NC-SA 4.0

A phenotype that can be scored

Read the network

- Tubular mitochondria form elongated, connected structures.
- Fragmented mitochondria appear as short, separate objects.

Griparic et al., JCB (2007), Results and Fig. 3

99%

Cells with fragmented mitochondria after OPA1 siRNA.

Re-expression connects OPA1 to fusion

- OPA1-null mouse fibroblasts have fragmented mitochondria.
- Fusion-active OPA1 splice forms restore tubular networks.
- A cell-hybrid assay measures recovery of mitochondrial fusion.

Depletion, imaging and rescue build the biological explanation.

Song et al., JCB (2007), Fig. 1C,D

Workflow 1: follow a gene from the papers to JUMP

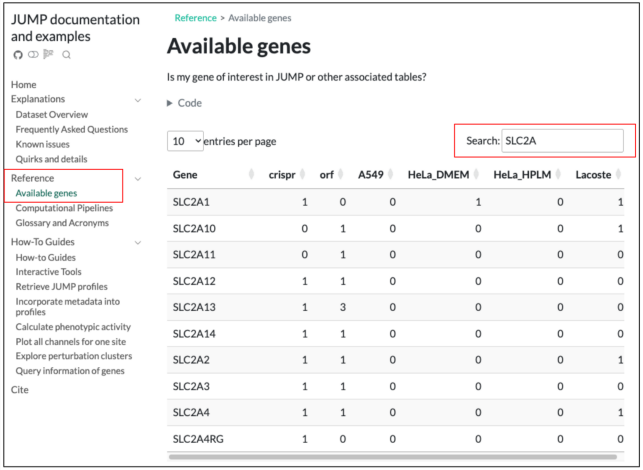
Search: OPA1

Perturbation	Available in JUMP
CRISPR knockout	Yes
ORF overexpression	Yes

- Follow the gene to its perturbation identifiers.
- Open the published feature, match and image tables.
- Use the Hub's worked examples to follow each step.

JUMP Hub gene availability; coverage reference

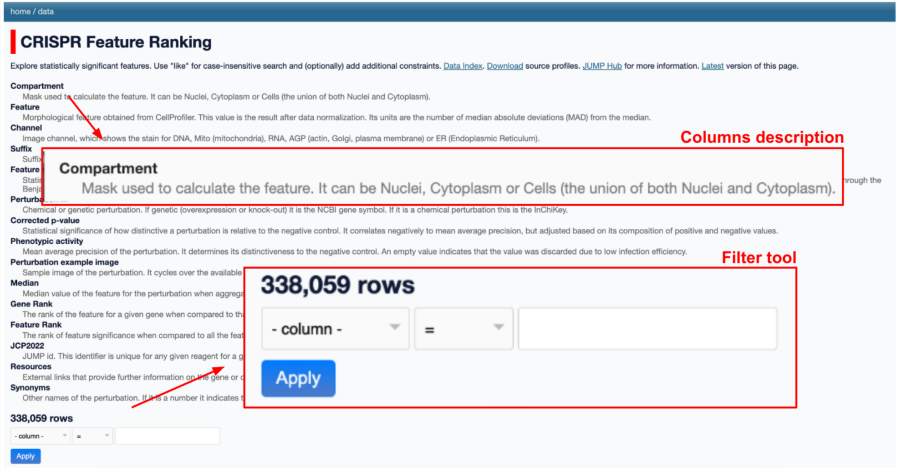
Gene availability: search a gene family



Gene	crispr	orf	A549	HeLa_DMEM	HeLa_HPLM	Lacoste
SLC2A1	1	0	0	1	0	1
SLC2A10	0	1	0	0	0	1
SLC2A11	0	1	0	0	0	0
SLC2A12	1	1	0	0	0	0
SLC2A13	1	3	0	0	0	0
SLC2A14	1	1	0	0	0	0
SLC2A2	1	1	0	0	0	1
SLC2A3	1	1	0	0	0	0
SLC2A4	1	1	0	0	0	1
SLC2A4RG	1	0	0	0	0	0

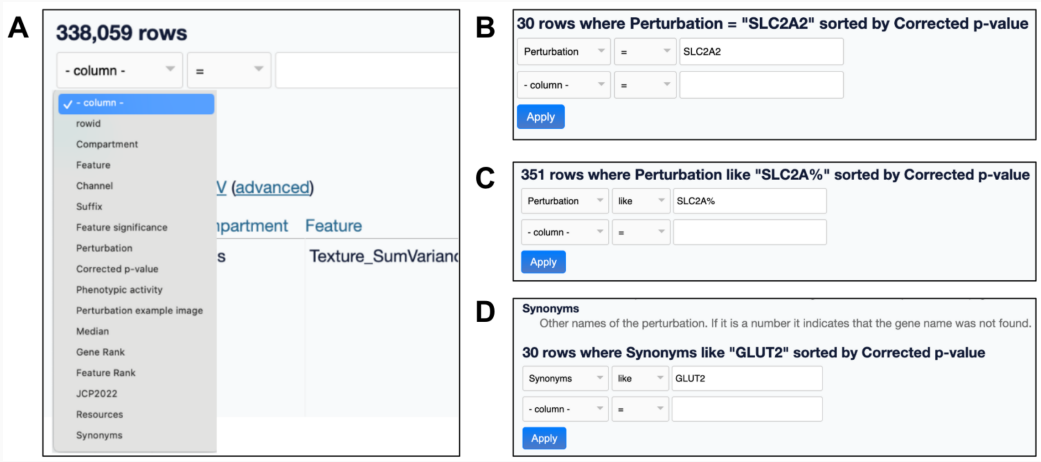
Read the dataset columns: find the available CRISPR and ORF perturbations.

Read the column definitions first



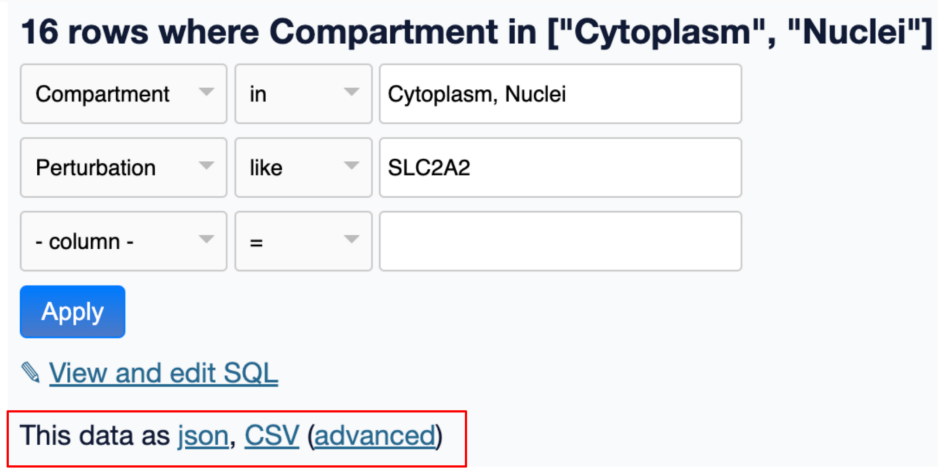
Use the red boxes: read the field definitions, then choose the filter for your question.

Filtering: exact gene, family and synonym



Compare the panels: select SLC2A2, expand to the SLC2A family, or search the GLUT2 synonym.

Keep the relevant compartments and save the view



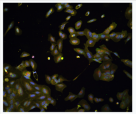
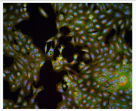
Save the query context: selected gene, compartments, identifiers and source link.

Workflow 2: read the returned phenotype

1. Read overall activity and the changing features.
2. Compare CRISPR and overexpression.
3. Follow similar profiles to their annotations and images.
4. Bring in controls from the same source and plates.

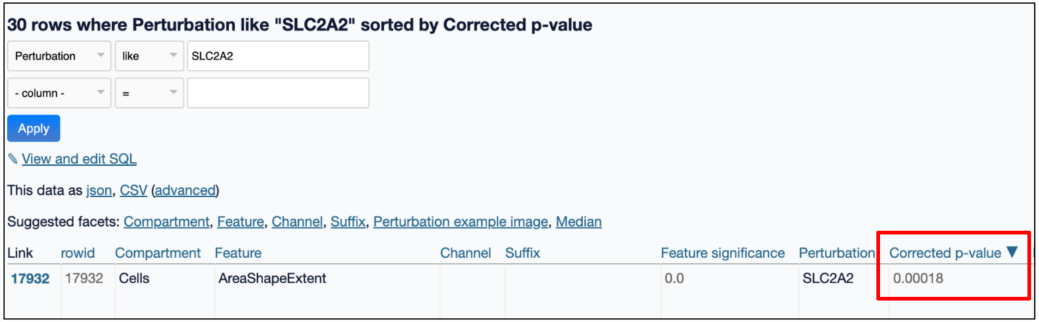
Worked examples: SLC2A2 for the tables; MYT1 for the image gallery.

Understand a feature-table result

Compartment	Feature	Channel ▲	Suffix	Feature significance	Perturbation	Corrected p-value	Phenotypic activity	Perturbation example image	Median	Gene Rank	Feature Rank
Cells	Intensity_MassDisplacement	RNA		1e-05	SNRNP200	0.00018	1.0		-0.965	999999	27
Nuclei	Texture_AngularSecondMoment	RNA	_5_03_256	0.0	CECR1	0.00018	0.77241		-0.203	999999	22

Read across the row: perturbation, feature, compartment, channel and statistical summary.

Overall activity: SLC2A2 CRISPR



30 rows where Perturbation like "SLC2A2" sorted by Corrected p-value

Perturbation | like | SLC2A2

- column - | = |

Apply

View and edit SQL

This data as json, CSV (advanced)

Suggested facets: Compartment, Feature, Channel, Suffix, Perturbation example image, Median

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Perturbation	Corrected p-value ▼
17932	17932	Cells	AreaShapeExtent			0.0	SLC2A2	0.00018

Read the boxed corrected p-value: it summarizes overall perturbation activity.

Describe the changes: SLC2A2 CRISPR

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Perturbation
17932	17932	Cells	AreaShapeExtent			0.0	SLC2A2
21804	21804	Nuclei	NeighborsSecondClosestDistance		_1	0.0	SLC2A2
22777	22777	Cells	NeighborsPercentTouching		_Adjacent	0.0	SLC2A2
33343	33343	Cytoplasm	RadialDistributionFracAtD		_mito_tubeness_12of20	0.0	SLC2A2

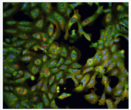
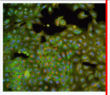
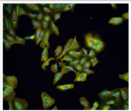
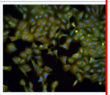
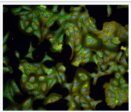
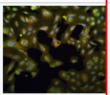
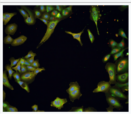
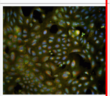
Read feature significance, rank and effect size to describe the morphological changes.

Compare perturbation types: SLC2A2 ORF

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance ▼
394755	394755	Cytoplasm	AreaShapeMinorAxisLength			8e-05
293369	293369	Cells	AreaShapeMinorAxisLength			0.00011
126201	126201	Cells	AreaShapeMinFeretDiameter			0.00036
292792	292792	Cytoplasm	AreaShapeMinFeretDiameter			0.00036

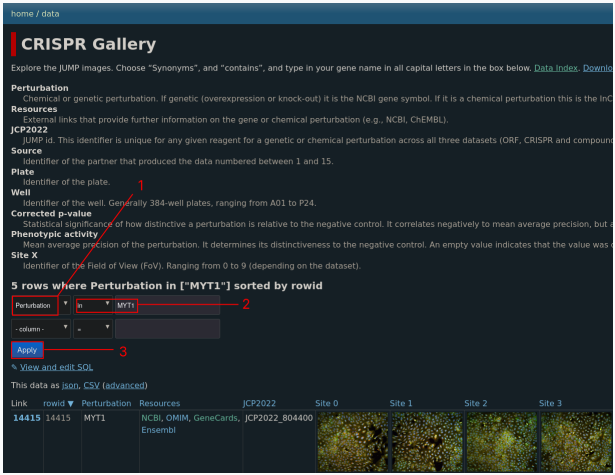
Compare the two levels: overall perturbation activity and the individual feature results.

Explore related profiles: SLC2A2 CRISPR

Link	rowid	Perturbation	Match	Perturbation example image	Match example image	Perturbation-Match Similarity ▲	Match resources
210190	210190	SLC2A2	RAD51			0.621	NCBI, OMIM, GeneCards, Ensembl
316101	316101	SLC2A2	BACE1			0.614	NCBI, OMIM, GeneCards, Ensembl
163302	163302	SLC2A2	FANCM			0.61	NCBI, OMIM, GeneCards, Ensembl
182659	182659	SLC2A2	TONSL			0.61	NCBI, OMIM, GeneCards, Ensembl

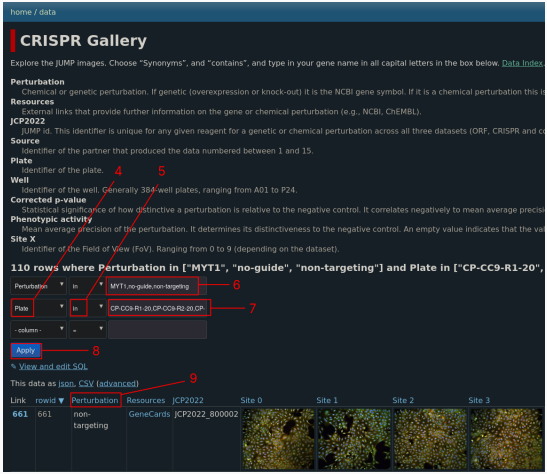
Sort by signed similarity, then follow the annotation links and inspect the example images.

Find the perturbed images: MYT1



Follow steps 1–3: select the perturbation, apply the filter, and identify its source and plates.

Bring in matched controls: MYT1



Follow steps 4–9: select the same plates and compare cell appearance, density, staining and focus.

Bring the workflow together with an agent

Start with the papers

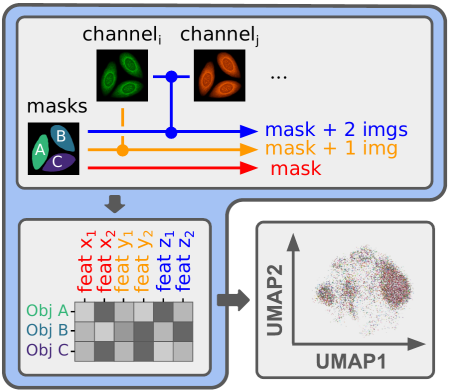
Find primary studies linking OPA1 depletion to mitochondrial fragmentation. Show the control and perturbed images, then summarize the genetic perturbation, imaging readout and main finding.

Follow the gene into JUMP:

“Find OPA1 in the Hub. Link the availability, feature, match and image resources, then explain how to read each view using the tutorial examples.”

Output: a cited biological summary and a linked route through the data.

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

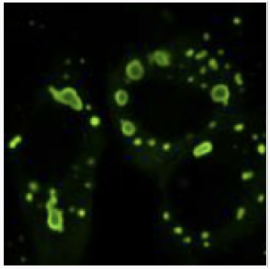
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

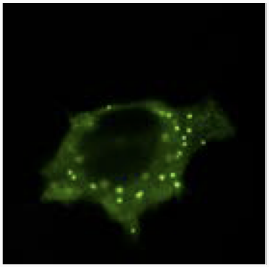
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

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Anne E. Carpenter
Shantanu Singh

Cimini Lab
Beth Cimini
Paula Llanos

Carpenter-Shen Lab
Anne E. Carpenter
Runxi Shen

JUMP partners



JUMP supporting partners



JUMP collaborators

Scientific:
Horizon Discovery,
a PerkinElmer Company
Administrative:
Pistoia Alliance
Academic:
Umeå University
Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

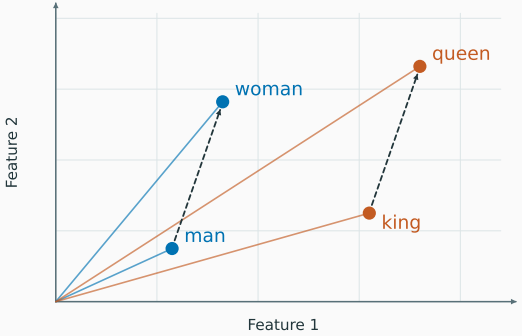
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

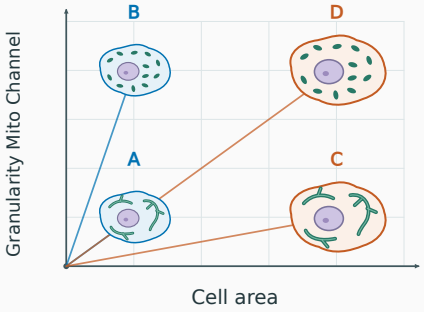
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

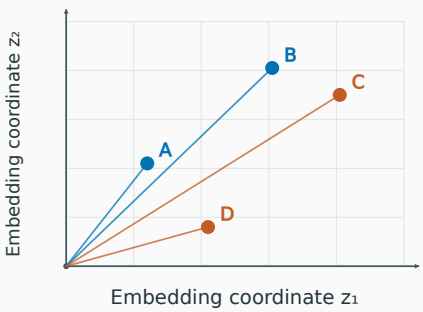


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



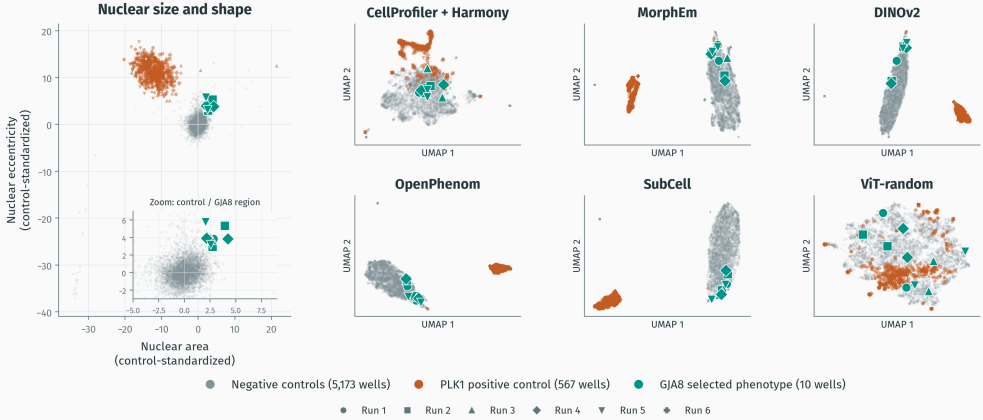
Control

Perturbation

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

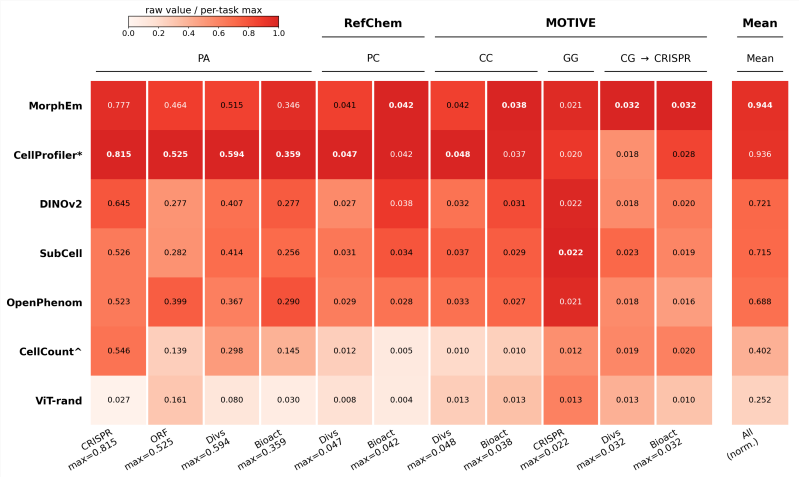
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

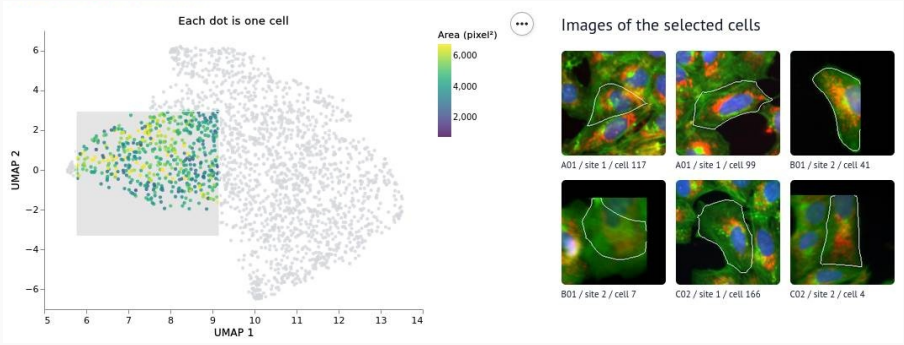


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

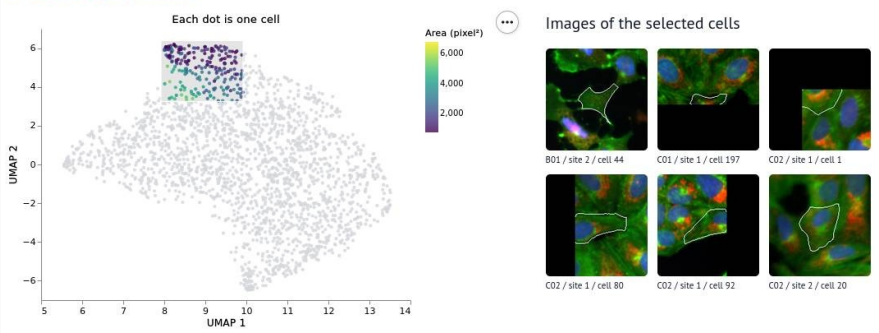
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the compound queries; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording